

Exercise 2

Estimating the effects of dynamic treatment strategies

Learning Objectives

By the end of this Exercise, you will be able to use inverse-probability (IP) weighting to appropriately estimate the causal effects of dynamic treatment strategies, generate adjusted absolute risks from real-world data, and interpret your findings.

Specifying the Target Trial

Imagine we are interested in estimating the effects of SGLT2i and GLP1-RA medication strategies. Because there is increasing interest in repurposing these medications as therapies for cancer patients, we will compare the effects of these agents on the 4-year risk of major adverse cardiovascular events (MACE) among individuals recently diagnosed with cancer.

We will make our causal question explicit by specifying the protocol of the target trial that would answer that question (**Table 1**).

Table 1. Specification of target trial	
Eligibility criteria	• Type II diabetes patients aged ≥ 30 years old with a diagnosis of non-metastatic cancer within the past 5 years • Fasting HbA1c of 7-11% within the past year • $eGFR \geq 45$ ml/min/1.73m² • ≥ 3 months of prior treatment with metformin • No prior history of any second-line diabetes treatments

	- No contraindication to SGLT2i or GLP1-RA, defined as a history of medullary thyroid cancer, urinary tract infections, pancreatitis, diabetic ketoacidosis, Fournier's gangrene, or severe kidney disease (defined as eGFR < 30 ml/min/1.73m², initiation of renal replacement therapy, or receipt of dialysis) - Enrolled in healthcare system for at least 12 months
Treatment strategies	SGLT2i: Initiation of SGLT2i with continued use throughout follow-up unless changes are clinically warranted GLP1: Initiation of GLP1-RA with continued use throughout follow-up unless changes are clinically warranted Under all treatment strategies, changes are considered clinically warranted and are left to patient and physician discretion if a patient develops any contraindication to SGLT2i or GLP1-RA (as defined above). At all other times, individuals may not discontinue, switch, or be simultaneously treated with an SGLT2i and a GLP1-RA.
Assignment procedures	Individuals are randomly assigned to one strategy at baseline, with both patients and treating physicians aware of treatment assignment.
Outcomes	Major adverse cardiovascular events (MACE), defined as death due to any cause, non-fatal myocardial infarction, or non-fatal stroke
Follow-up	Eligible patients are followed from baseline until the earliest of MACE, 4 years, or administrative end of follow-up (July 1, 2023)
Causal contrasts	Intention-to-treat and per-protocol effects
Statistical analysis	<i>Intention-to-treat</i> Risks estimated using pooled logistic regression. <i>Per-protocol</i> Same as the intention-to-treat analysis except that individuals are censored when they deviate from their assigned treatment strategy and IP weights are applied to adjust for prognostic factors related to adherence.
eGFR: estimated glomerular filtration rate; SGLT2i: sodium-glucose cotransporter 2 inhibitors; GLP1-RA: glucagon-like peptide 1 receptor agonists. For ease of illustration, we consider a simplified version of the strategies described during lecture.	

For ease of illustration, we assume no loss to follow-up, consider an outcome that is not subject to competing events, and do not incorporate a grace period or allowable gap for

treatment to be considered continuous. However, the methods we describe can easily be extended to handle these additional features.

Emulating the Target Trial Using Observational Data

Conducting this target trial would require recruiting thousands of cancer patients, randomizing them to each treatment strategy, and then carefully following them for many years. Thus, given the time constraints of this workshop and the need to produce timely evidence to inform decision-making today, we will *emulate* this target trial using observational, real-world data based on The oBservational Evaluation of Second line Therapy MEDications in Diabetes (BESTMED) Consortium.

The oBservational Evaluation of Second line Therapy MEDications in Diabetes (BESTMED) Consortium

BESTMED is a consortium which draws from large healthcare institutions and health insurance companies. It has a combined patient population of over 130 million individuals across the United States (US), with data spanning from January 1, 2014, to July 31, 2023. The dataset includes longitudinal electronic health records and health insurance claims data from patients with type 2 diabetes who have been on metformin for at least three months and are at moderate risk of cardiovascular disease. Additional details on BESTMED can be found at <https://bestmed.org/>.

Simulated data

Because the BESTMED consortium contains real patient data, sharing is restricted. Therefore, we created a simulated dataset (based on the relationships between a subset of the variables recorded in BESTMED) which will be used throughout Exercise 2. These data are in “long format” where each row corresponds to one person-month under follow-up. The simulated dataset has already been restricted to individuals who met the eligibility criteria for the target trial and initiated one of the two treatment strategies of interest (**Table 1**). The data dictionary for the simulated dataset is shown below (**Table 2**).

Table 2. Data dictionary for the simulated dataset.

Variable	Name	Type	Coding
<i>Time-fixed variables</i>			
Participant ID	id	Numerical	Unique numerical number
Assigned strategy	strategy	Binary	0: SGLT2i 1: GLP1
Age	Age	Numerical	Years, Z-score transformed
Female sex assigned at birth	Female	Binary	0: No 1: Yes
Social Deprivation Index	SDI	Numerical	See here
Medical insurance	Insurance	Binary	0: No 1: Yes
<i>Time-varying variables</i>			
Follow-up time in months	Time	Numerical	Months
GLP1 use in month k	glp1	Binary	0: No 1: Yes
GLP1 use in month k-1	lag1_glp1	Binary	0: No 1: Yes
SGLT2i use in month k	sglt2i	Binary	0: No 1: Yes
SGLT2i use in month k-1	lag1_sglt2i	Binary	0: No 1: Yes
eGFR	eGFR	Numerical	ml/min/1.73m ²
Obesity	Obesity	Categorical	1: Underweight 2: Normal weight 3: Overweight 4: Obesity Class 1&2

			5: Obesity Class 3
HbA1c	hba1c	Numerical	%
Developed a contraindication to SGLT2i or GLP1 (see Table 1)	excused	Binary	0: No 1: Yes
MACE	mace	Binary	0: No 1: Yes

We are now ready to begin estimating the causal effects of dynamic treatment strategies using g-methods! We will use the estimation procedure described in lecture. In Data Analysis Part 1 we will create time-varying IP weights for the dynamic treatment strategies outlined in Table 1. Then, in Data Analysis Part 2, we will define and implement censoring rules to create a dataset in which all eligible individuals follow their “assigned” dynamic treatment strategy throughout the follow-up. Finally, in Data Analysis Part 3, we will estimate the causal effect of interest via a dynamic marginal structural model.

Data Analysis Part 1: IP Weights

- a) Fit models to estimate the denominator probabilities for the non-stabilized IP weights, including (i) a logistic regression model fit at time 0 to estimate baseline probabilities and (ii) two pooled logistic regression models at times > 0 to estimate the time-specific probabilities after baseline, separately among individuals who initiated an SGLT2i vs. a GLP1-RA at baseline. These latter models are referred to as pooled logistic regression models because they are fit to a long-format dataset where information can be pooled across multiple time points.

See “Exercise 2_Code.R” and “Exercise 2_Code.sas”

- b) Which observations contribute to the models you fit in Part a? What do you think would happen if all observations were allowed to contribute to these models?

In the first model, all observations at the baseline time contribute. No one is non-adherent or excused at baseline.

In the second two models, only post-baseline adherent observations that were not previously excused (i.e., developed a contraindication to SGLT2i or GLP1) are included in each model. If excused observations were included, then our estimates could be biased due to violations of the sequential positivity assumption (because people who develop a contraindication may no longer have a positive probability of continuing their medication). We chose to only include adherent observations (who had not previously deviated from their assigned strategy) and to use separate models for individuals who initiated an SGLT2i vs. a GLP1-RA at baseline as modeling decisions that we could modify, if desired.

- c) Take the cumulative product of the time-specific probabilities that you estimated in Part a) and construct time-varying IP weights for each eligible individual. Whenever an individual is excused, assign them a time-specific weight contribution of 1.

See “Exercise 2_Code.R” and “Exercise 2_Code.sas”

Data Analysis Part 2: Censoring

- a) Based on the dynamic treatment strategies described in Table 1, define censoring rules for each treatment strategy that state when an individual would be censored due to non-adherence.

SGLT2i: Individuals are censored if and when they discontinue, switch, or are simultaneously treated with an SGLT2i and a GLP1-RA AND they were NOT previously excused (i.e., they had NOT previously developed a contraindication to SGLT2i or GLP1-RA).

GLP1-RA: Individuals are censored if and when they discontinue, switch, or are simultaneously treated with an SGLT2i and a GLP1-RA AND they were NOT previously excused (i.e., they had NOT previously developed a contraindication to SGLT2i or GLP1-RA).

- b) Implement the censoring rules you defined in Part a) to create a dataset in which all eligible individuals follow their “assigned” treatment strategy throughout the follow-up.

See “Exercise 2_Code.R” and “Exercise 2_Code.sas”

- c) Now that you have created the censored dataset, use this dataset to truncate the IP weights at their 99th percentile. Describe the distribution of the truncated weights.

Minimum: 1.174

1st quartile: 1.756

Median: 2.151

Mean: 2.496

3rd quartile: 2.831

Maximum: 8.022

Data Analysis Part 3: Estimating Effects via a Dynamic Marginal Structural Model

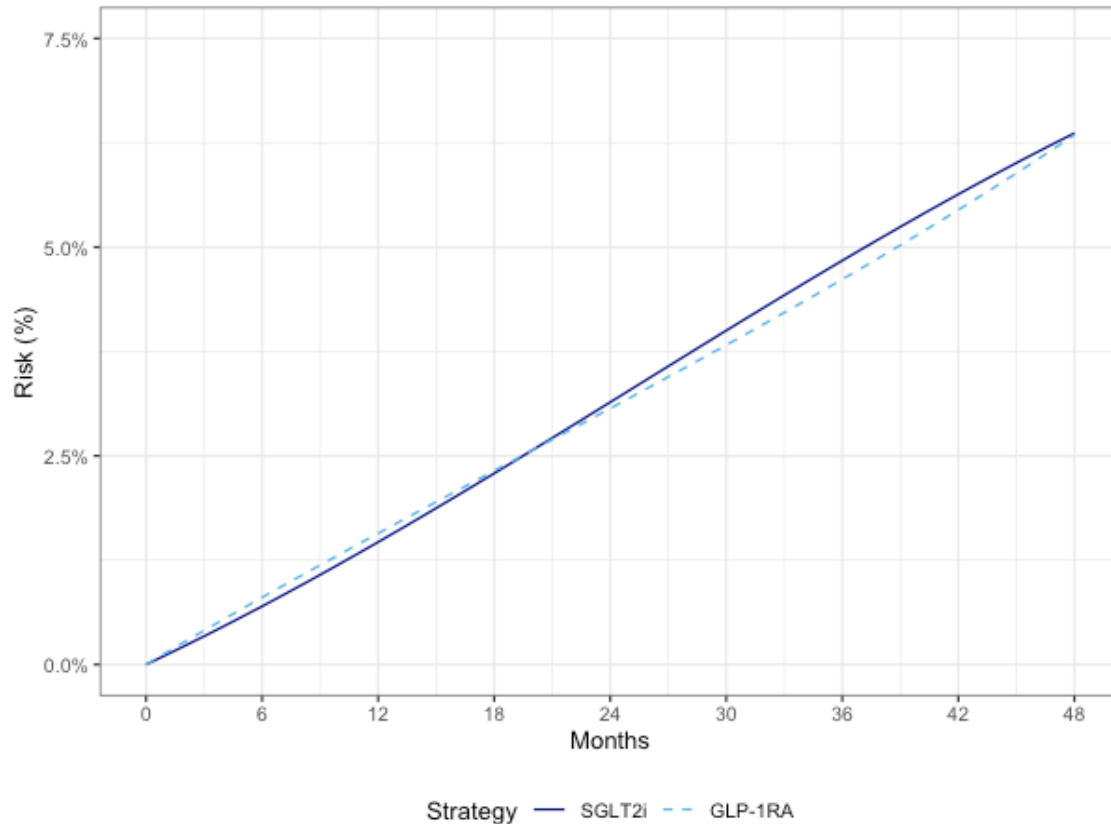
- a) Using the IP weighted and censored dataset that you created in Parts 1 and 2, fit a pooled logistic regression model for the outcome of MACE. Recall from Lecture 2 that the parameters from this model will approximate the parameters of a dynamic marginal structural model and will correspond to causal effects when necessary assumptions are met.

See “Exercise 2_Code.R” and “Exercise 2_Code.sas”

- b) Using the parameter estimates from the model you fit in part a), estimate the marginal risk of MACE at each month of follow-up (from baseline through the end of follow-up at 4 years) under each dynamic treatment strategy.

See “Exercise 2_Code.R” and “Exercise 2_Code.sas”

c) Using the risks you estimated in part b), construct adjusted risk curves for the outcome of MACE under each treatment strategy.



d) Using the risks you estimated in part b), estimate the 4-year risk difference and risk ratio for MACE. Interpret each of these measures in the context of your study (see Table 1).

- **Risk ratio:** The estimated risk of MACE after 4 years had everyone initiated and continuously taken a GLP1-RA over follow-up *unless changes were clinically warranted* is 1.00 times that had everyone initiated and continuously taken an SGLT2i over follow-up *unless changes were clinically warranted*.
- **Risk difference:** The excess cases of MACE after 4 years had everyone initiated and continuously taken a GLP1-RA over follow-up *unless changes were clinically warranted* compared to had everyone initiated and continuously taken an SGLT2i over follow-up *unless changes were clinically warranted* is 0 per 100 persons.

Note that, for simplicity of illustration, we did not generate confidence intervals for the risk ratio or the risk difference. In practice, confidence intervals are usually estimated using bootstrapping.

Summary

In this Exercise, we learned how to estimate the effects of dynamic treatment strategies using IP weighting and censoring of a dynamic marginal structural model. Alternative estimators can also be used to estimate these effects (e.g., the g-formula), as discussed in Lecture 2.